

Tracking

Tracking

3D Human Pose/Motion Tracking

Paradigm	Complexity	Number	Modeling	Motion
Reconstruction	Body (MHR) + Camera	1 / N person	Geometry	Animation
+ Generation (observation, text, audio, ...)	+ Interaction (objects /scenes)	1 / N objects	+ Appearance (texture, materials, lighting)	Motion generation
+ Freeview-point video generation	+ Cloth		+ Functionality (semantic, parts)	Motion Transfer (human2human, human2robot, human2animal)

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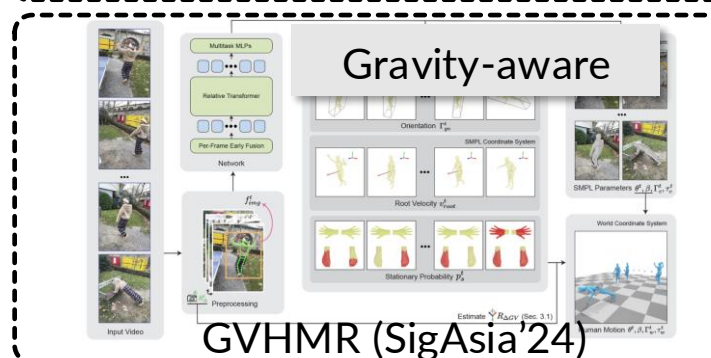
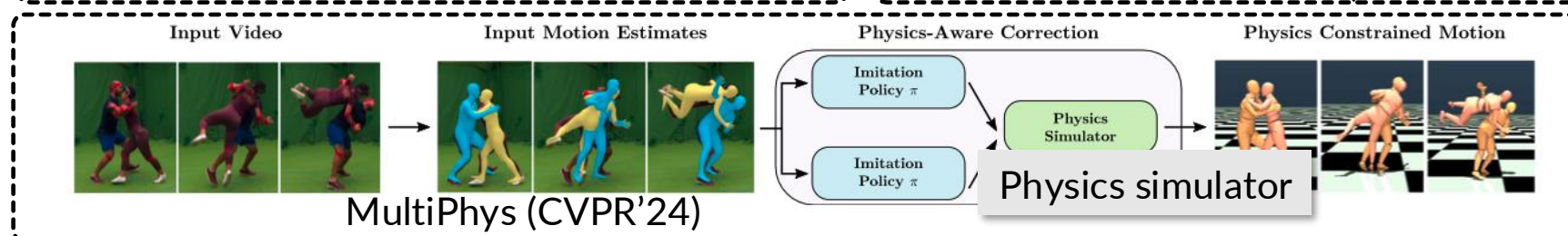
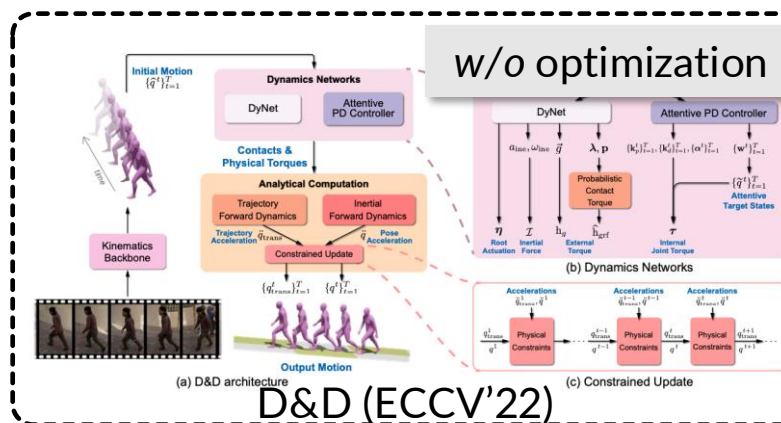
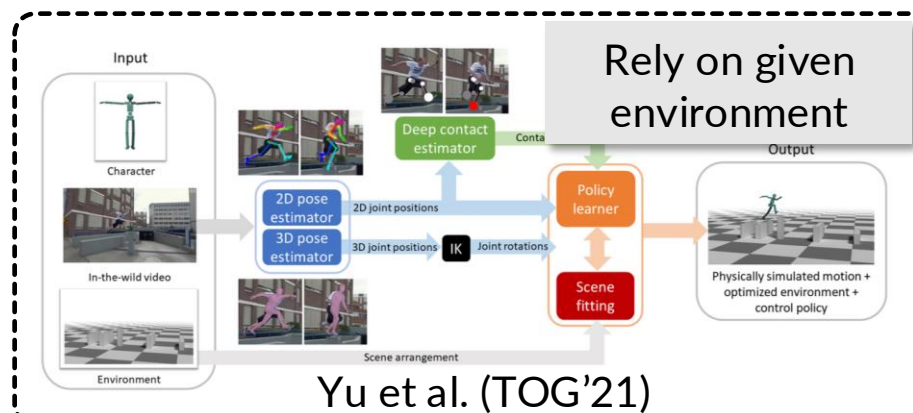
- Local Human Mesh Reconstruction/Tracking (in camera coords.)
 - Single image/Monocular video
 - Single-person/Multi-person
 - Deterministic/Probabilistic
- Global Human Motion Tracking (in world coords.)
 - Optimization/ Regression
 - Physics-aware
- Joint Human-Scene Tracking
 - Global human motion estimation
 - Generic scene reconstruction
 - Interaction (scene/objects)
 - Novel view synthesis, Generation (animation, motion transfer)
 - Physics-aware

Tracking

3D Human Pose/Motion Tracking

- Global Human Motion Tracking (in world coords.)

Method	Venue	Note
Yu et al	TOG'21	Physics
GLAMR	CVPR'22	
D&D	ECCV'22	Physics
TRACE	CVPR'23	
SLAHMR	CVPR'23	Optimization
PACE	3DV'24	Optimization
WHAM	CVPR'24	
MultiPhys	CVPR'24	Physics
COIN	ECCV'24	
TRAM	ECCV'24	
WHAC	ECCV'24	
GVHMR	SigAsia'24	Gravity
GENMO	ICCV'25	Multitask
CRISP	arxiv'25	Physics



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WHAC	ECCV'24	
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GENMO	ICCV'25	Multitask
CRISP	arxiv'25	Physics

- Possible directions (**w/o scene**)
 - SAM 3D Body Tracking
 - Multi-person (occl mask, human body contact, relative depth)
 - Temporal cues (global tracking in *world* coords.)
 - Optimization/Feed-forward
 - Physics-aware (terrain, ground contact...) / Simulation-ready



Frame-by-frame estimates from SAM 3D Body

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Method	Venue	Note
HOSNeRF	ICCV'23	opt; dynamic NeRF
DUST3R	CVPR'24	ff; 3D GT; static (2 images)
VGGT	CVPR'25	ff; 3D GT; static
CUT3R	CVPR'25	ff; 3D GT; dynamic online
HSfM	CVPR'25	opt; static; single-frame, <u>multiview</u>
HAMSt3R	ICCV'25	ff; static; single-frame, <u>multiview</u>
<u>Easi3R</u>	ICCV'25	training-free; dynamic; (ext DUST3R)
Uni3C	SigAsia'25	video generation
JOSH3R	<u>arxiv'25</u>	opt; dynamic; (submit to ICLR26, 4666)
Human3R	<u>arxiv'25</u>	ff; dynamic online; (CUT3R+SMPL)
DA3	<u>arxiv'25</u>	ff; 3D GT; dynamic; (vs. VGGT)
SHARE	<u>arxiv'25</u>	opt; dynamic

Tracking

HOSNeRF (ICCV'23)

Input: a monocular video + calibrated pose

Output: 4D dynamic human-object-scene NeRF

- Estimated SMPL (ROMP), human+object mask (MaskRCNN)
- Three-stage training
- Per-video training (5 days, 4 V100)

HOSNeRF: Dynamic Human-Object-Scene Neural Radiance Fields from a Single Video

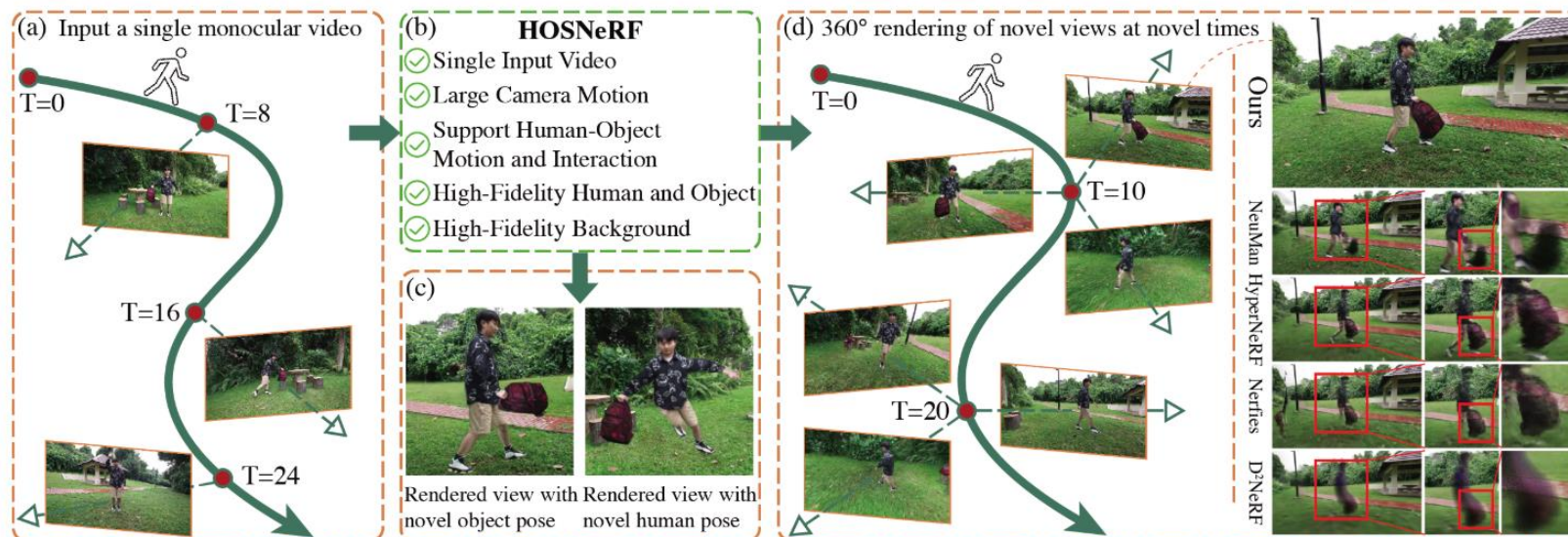
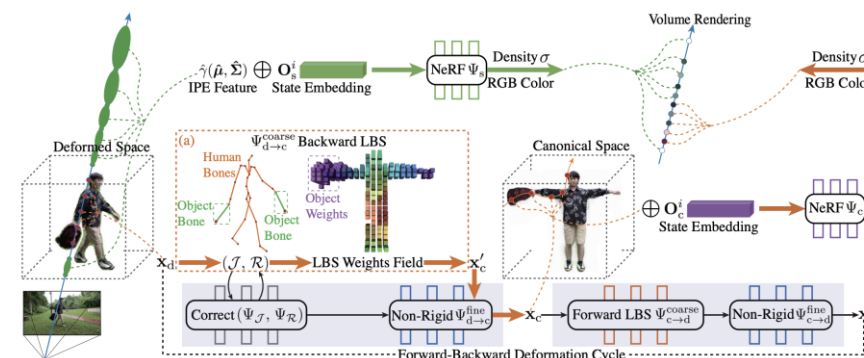
ICCV 2023

Jia-Wei Liu^{1,2}, Yan-Pei Cao², Tianyuan Yang¹, Zhongcong Xu¹, Jussi Keppo^{4,5},

Ying Shan², Xiaohu Qie³, Mike Zheng Shou¹

¹Show Lab, National University of Singapore ²ARC Lab, ³Tencent PCG

⁴Business School, ⁵Institute of Operations Research and Analytics, National University of Singapore



360° Free-Viewpoint Renderings from HOSNeRF – **BACKPACK**



Tracking

VGGT (CVPR'25)

VGGT: Visual Geometry Grounded Transformer

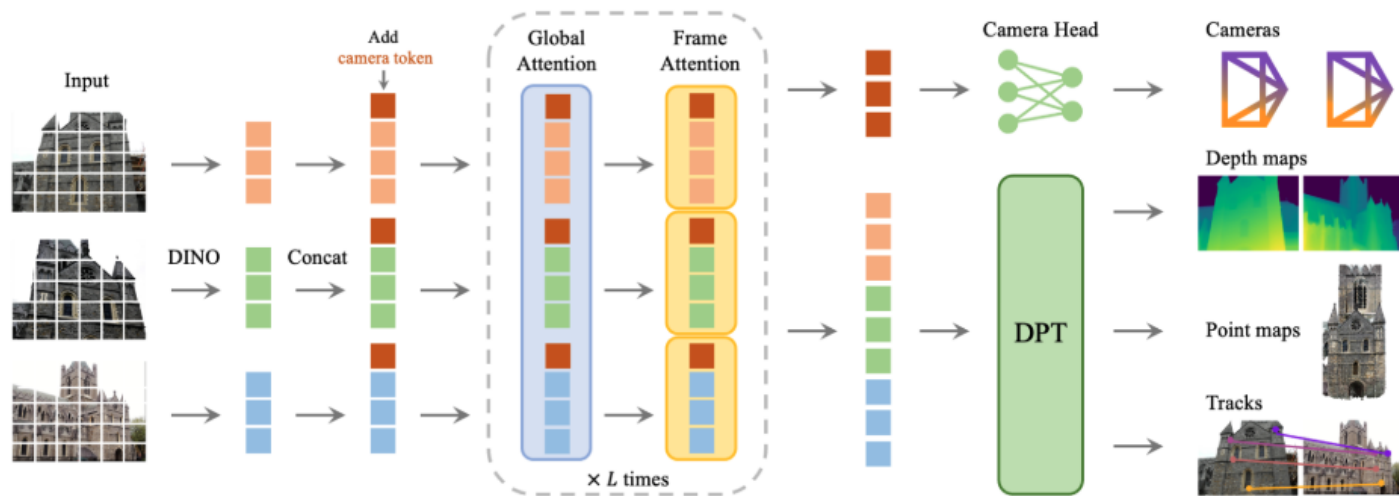
Jianyuan Wang^{1,2}, Minghao Chen^{1,2}, Nikita Karaev^{1,2}

Andrea Vedaldi^{1,2}, Christian Rupprecht¹, David Novotny²

¹Visual Geometry Group, University of Oxford, ²Meta AI

CVPR 2025 Best Paper Award

<https://huggingface.co/spaces/facebook/vggt>



1. **Static scene**
2. Multiple input types
 1. Video frames
 2. Photo collection
3. Unknown cameras
4. Need 3D GT
5. Feed-forward
 1. Large-scale training
 2. Time & GPU-consuming



Input video



VGGT using 1 frame



VGGT using all frames

Tracking

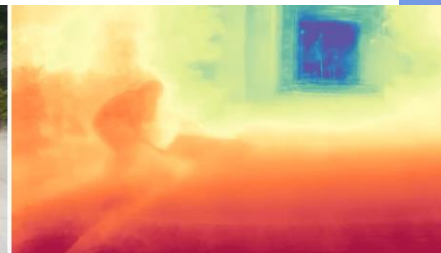
DepthAnything3 (a)

Recovers geometry & 3DGS from

*Depth-ray representation

Input: any number of images (unknown cameras)

Output: depth, raymaps (6: ray ori, ray dir), pointmaps, 3DGS

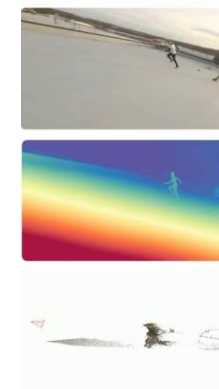
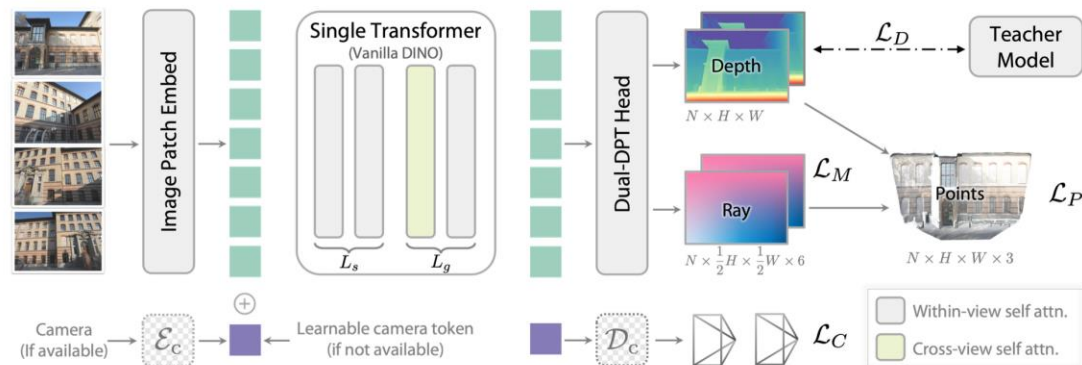


Depth Anything 3

Recovering the Visual Space from Any Views

Yongqiang Lin*, Sili Chen*, Jun Hao Liew*, Donny Y. Chen*, Zhenyu Li, Guang Shi, Jiashi Feng, Bingyi Kang**

*Project Lead, **Equal Contribution



Official demo



Input video



DA3 using all frames

DA3 3DGS rendering

1. Use depth-ray representation
 1. Avoid directly estimate camera extrinsic
 2. Estimated raymap -> camera poses
2. +3D GS (feed-forward)
3. Large compute cost (128 H100, 200k iters)

Tracking CUT3R (CVPR'25)

<https://cut3r.github.io/>

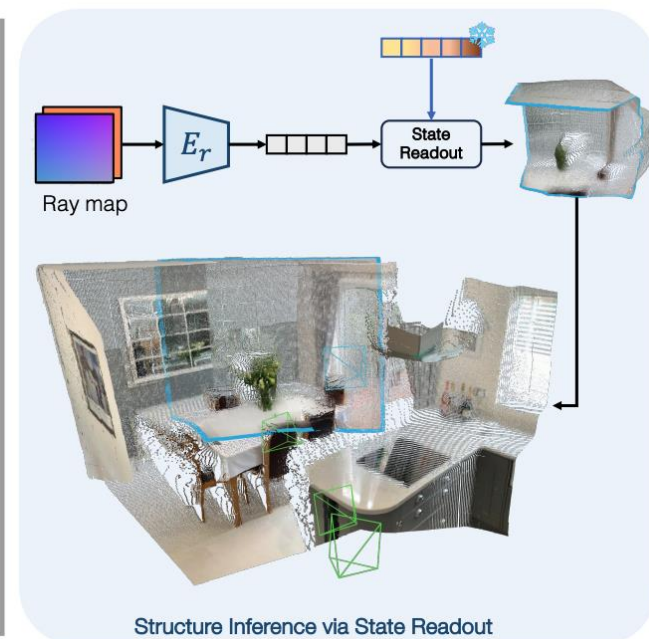
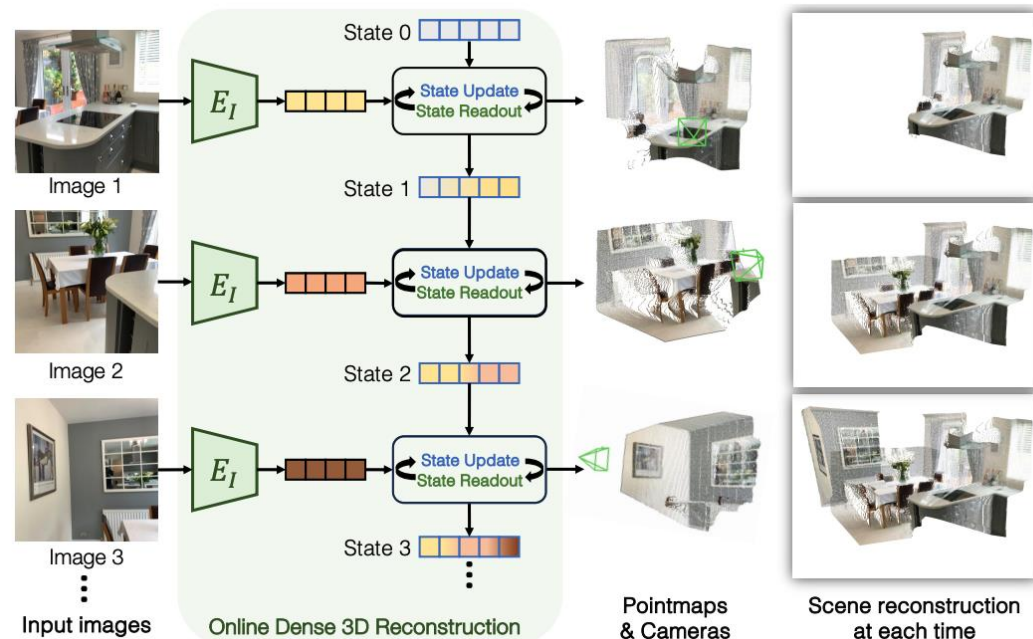
Continuous 3D Perception Model with Persistent State

Qianqian Wang^{1,2*} Yifei Zhang^{1*} Aleksander Holynski^{1,2} Alexei A Efros¹ Angjoo Kanazawa¹

¹UC Berkeley ²Google DeepMind

(*: equal contribution)

CVPR 2025 (Oral)



Dynamic Scene
Reconstruction

3D Reconstruction (Video)

3D Reconstruction (Photo
Collection)

Inferring Unseen Structure

Dynamic Scene Reconstruction

Our method can reconstruct dynamic scenes in a feed-forward, online manner without per-video optimization.



0.9 1x



4.2 1x

Open in New Tab for Better Visualization

Open in New Tab for Better Visualization

1. Online updating (dynamic scene)
2. Multiple input types
 1. Video frames
 2. Photo collection
3. Unknown cameras
4. Need 3D GT
5. Feed-forward
 1. Large-scale training (32 datasets)
 2. Time & GPU-consuming

Dataset Name	Scene Type	Metric?
ARKitScenes [5]	Indoor	Yes
ARKitScenes-HighRes [5]	Indoor	Yes
ScanNet [19]	Indoor	Yes
ScanNet++ [121]	Indoor	Yes
TartanAir [108]	Mixed	Yes
Waymo [93]	Outdoor	Yes
MapFree [3]	Outdoor	Yes
BlendedMVS [120]	Mixed	No
HyperSim [75]	Indoor	Yes
MegaDepth [53]	Outdoor	No
Unreal4K [99]	Mixed	Yes
DL3DV [55]	Mixed	No
CO3Dv2 [74]	Object-Centric	No
WildRGBD [115]	Object-Centric	Yes
VirtualKITTI2 [13]	Outdoor	Yes
Matterport3D [15]	Indoor	Yes
BEDLAM [9]	Mixed	Yes
Dynamic Replica [43]	Indoor	Yes
PointOdyssey [130]	Mixed	Yes
Spring [60]	Mixed	Yes
MVS-Synth [40]	Outdoor	Yes
UASOL [6]	Outdoor	Yes
OmniObject3D [114]	Object-Centric	Yes
RealEstate10K [131]	Indoor	No
MVImgNet [123]	Object-Centric	No
CoP3D [84]	Object-Centric	No
EDEN [50]	Outdoor	Yes
IRS [105]	Indoor	Yes
Synscapes [111]	Outdoor	Yes
3D Ken Burns [67]	Mixed	No
SmartPortraits [48]	Indoor	Yes
UrbanSyn [32]	Outdoor	Yes
HOI4D [57]	Indoor	Yes

Tracking HSfM (CVPR'25)

Step 1: Input Processing and Feature Extraction

From the sparse input images, we extract 2D human joints using VIT-Pose and estimate 3D joint positions and body shape using HMR2. For scene and camera reconstruction, we use DUST3R, a state-of-the-art data-driven SfM method. We assume known re-identification of people across camera views.



Preprocessing with domain-specific neural nets



Keypoints from VITPose



Human Meshes from HMR 2.0



Cameras and Scene from DUST3R

Step 2: Resolving Scale Ambiguity

SfM methods often suffer from scale ambiguity. We address this by estimating camera parameters based on human body size and orientation. After alignment, people are roughly placed in a world of consistent scale.



Sparse set of images from uncalibrated cameras capturing a scene with multiple people.



Reconstructed 3D scene with camera poses, 3D human meshes and scene pointmaps

Step 3: Joint Human and Scene Reconstruction

We adapt the global alignment loss from DUST3R to jointly estimate humans and the scene. The HSfM loss consists of three terms: a keypoint re-projection term performing bundle adjustment on 2D body joints, a body shape regularizer, and the global alignment loss from DUST3R.

HSfM Loss:

$$\underbrace{\frac{1}{HC} \sum_{c=1}^C \sum_{h=1}^H \frac{1}{b_{2D}^{ch}} ||c_{2D}^{ch}(J_{2D}^{ch} - K^c(R^c J_{3D}^h + \alpha t^c))||_2}_{\text{Keypoint Loss (re-projection)}} + \underbrace{\frac{1}{H} ||\beta^h||_2 + \lambda_s}_{\text{Body Shape Regularizer}} \underbrace{\sum_{(c_p, c_j) \in \mathcal{E}} \sum_{i=1}^{HW} Q_i^{c_p, c_j} ||\mathcal{S}_i^{c_j} - \sigma^{c_p, c_j \rightarrow w} P_*^{c_p, c_j \rightarrow w} X_i^{c_p, c_j}||}_{\text{Scene Alignment}}$$

1. Static scene (Single-frame)
2. Need multiview images
3. Multi-stage
4. Post-optimization

Reconstructing People, Places, and Cameras

Lea Müller* Hongsuk Choi* Anthony Zhang Brent Yi Jitendra Malik Angjoo Kanazawa

UC Berkeley
*equal contribution

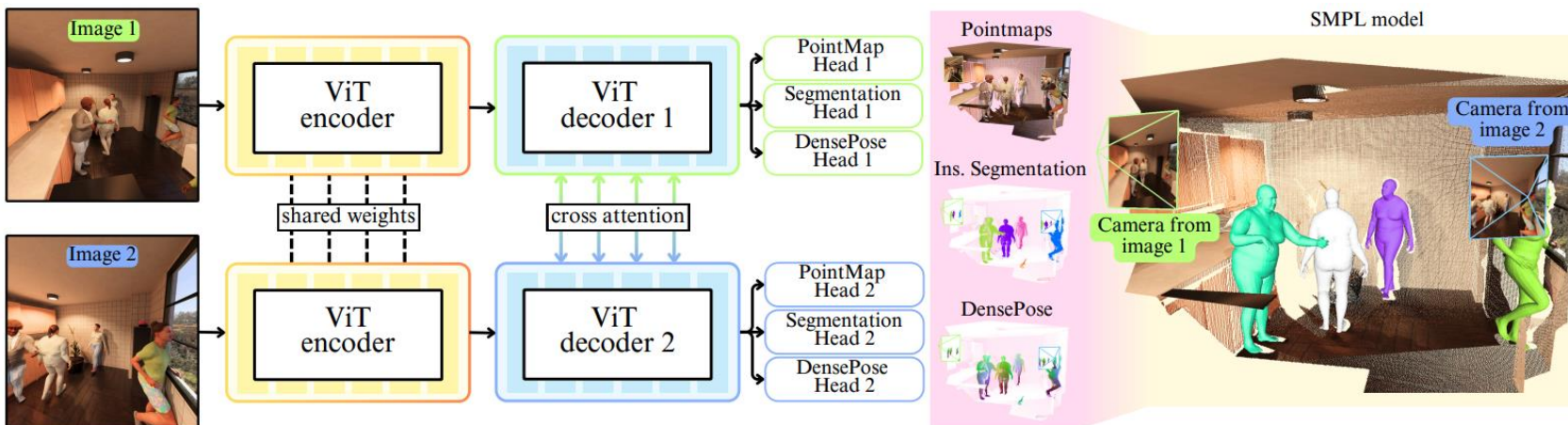
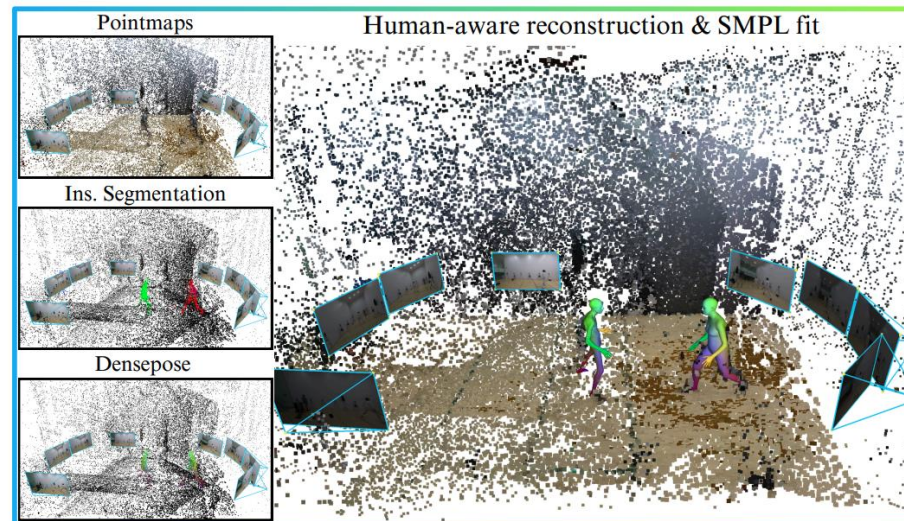
CVPR 2025 (Highlight)

Tracking

HAMSt3R (ICCV'25)

Table 2. Human-centric evaluation metrics on EgoHumans and EgoExo4D.

Method	EgoHumans			EgoExo4D	
	W-MPJPE ↓	GA-MPJPE ↓	PA-MPJPE ↓	W-MPJPE ↓	PA-MPJPE ↓
Multi-HMR [6]	7.66	0.99	0.12	2.88	0.07
UnCaliPose [66]	3.51	0.67	0.13	2.90	0.13
HSfM (init) [41]	4.28	0.51	0.06	5.29	0.07
HSfM [41]	1.04	0.21	0.05	0.56	0.06
HAMSt3R (Ours)	3.80	0.42	0.14	0.51	0.09



1. Static scene
2. Single-frame, multiview images
3. **First feed-forward**
* Except that SMPL is fitted

are fine-tuned. The overall training loss $\mathcal{L}$ is a weighted sum of the MAST3R loss $\mathcal{L}_{\text{MASt3R}}$, the segmentation loss $\mathcal{L}_{\text{seg}}$, the dense pose loss $\mathcal{L}_{\text{dp}}$, and the binary mask loss $\mathcal{L}_{\text{mask}}$:

$$\mathcal{L} = \mathcal{L}_{\text{MASt3R}} + \lambda_1 \mathcal{L}_{\text{seg}} + \lambda_2 \mathcal{L}_{\text{dp}} + \lambda_3 \mathcal{L}_{\text{mask}}, \quad (1)$$

Table 6. Multi-view depth evaluation (ScanNet) denote training on data from the same domain.

Method	KITTI		ScanNet		ETH3D		DTU		T&T		Average	
	rel. ↓	$\tau \uparrow$	rel. ↓	$\tau \uparrow$	rel. ↓	$\tau \uparrow$	rel. ↓	$\tau \uparrow$	rel. ↓	$\tau \uparrow$	rel. ↓	$\tau \uparrow$
DeepV2D (ScanNet) [61]	10.00	36.20	4.40	54.80	11.80	29.30	7.70	33.00	8.90	46.40	8.60	39.90
DUST3R [64]	5.88	47.67	3.01	72.54	3.04	75.17	2.92	73.94	2.93	78.51	3.56	69.56
MASt3R [33]	3.54	65.68	4.17	65.22	2.44	82.77	3.46	66.89	2.04	87.88	3.13	73.69
DUNE [51]	<u>4.88</u>	50.76	4.24	59.68	2.48	77.97	2.69	75.63	2.60	79.19	3.38	68.65
HAMSt3R (Ours)	5.60	45.66	4.43	56.50	2.96	71.68	5.31	57.62	3.01	73.53	4.26	61.00

HAMSt3R: Human-Aware Multi-view Stereo 3D Reconstruction

Sara Rojas^{1,2*} Matthieu Armando² Bernard Ghanem¹
Philippe Weinzaepfel² Vincent Leroy² Grégory Rogez²

¹KAUST ²NAVER LABS Europe

Tracking

Easi3R (ICCV'25)

A simple **training-free** approach adapting DUST3R for dynamic scenes.

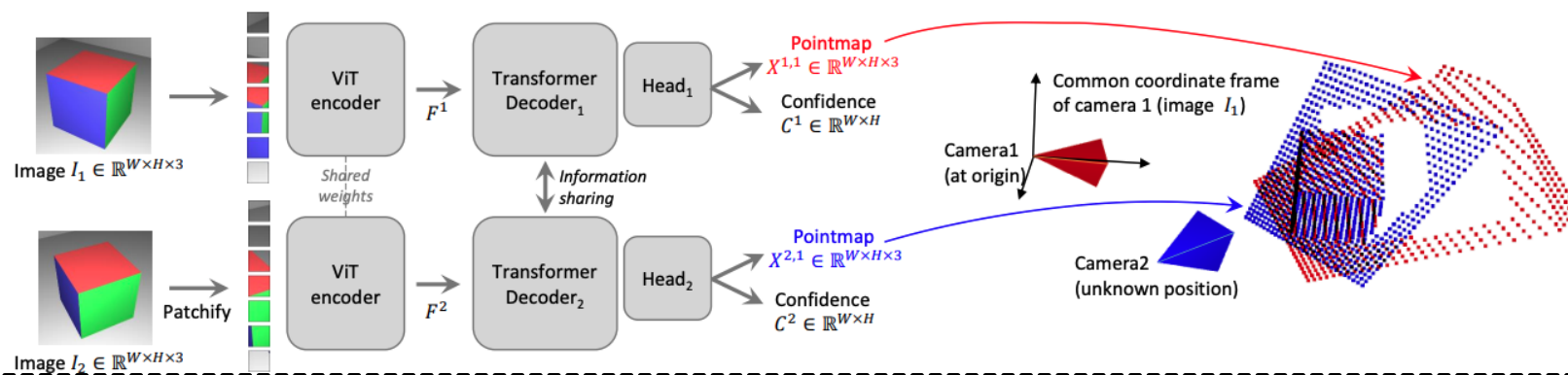
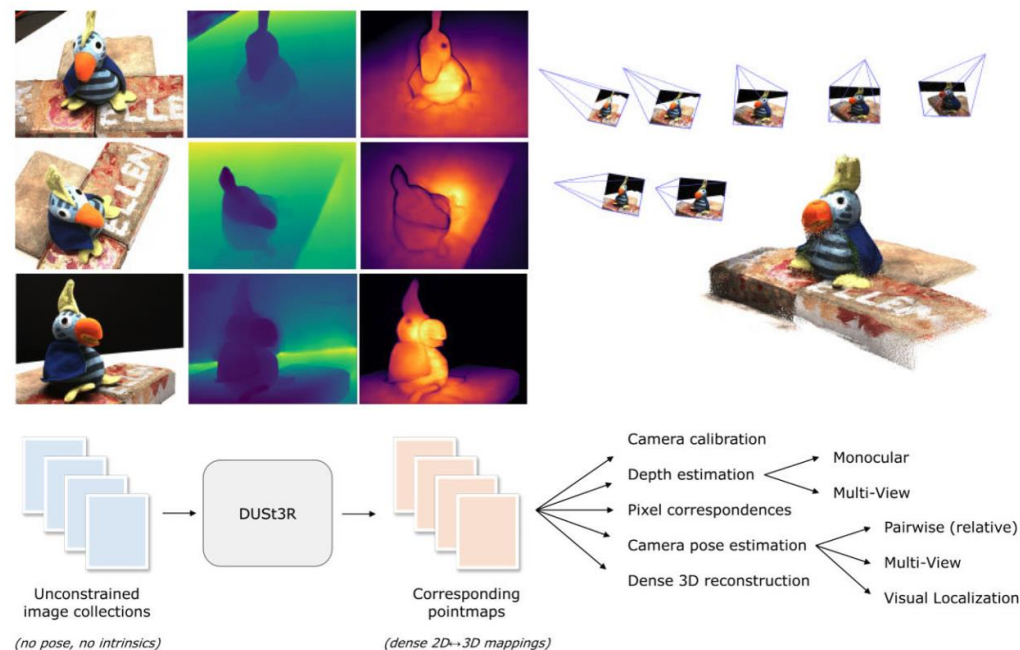
DUST3R: Geometric 3D Vision Made Easy

Shuzhe Wang*, Vincent Leroy†, Yann Cabon†, Boris Chidlovskii† and Jerome Revaud†
 *Aalto University †Naver Labs Europe

Input: an unconstrained image collection $I^1, I^2 \in \mathbb{R}^{W \times H \times 3}$

Output: a set of corresponding pointmaps $X^{1,1}, X^{2,1} \in \mathbb{R}^{W \times H \times 3}$

- Camera params
- Pixel correspondences
- Depthmaps
- 3D reconstruction



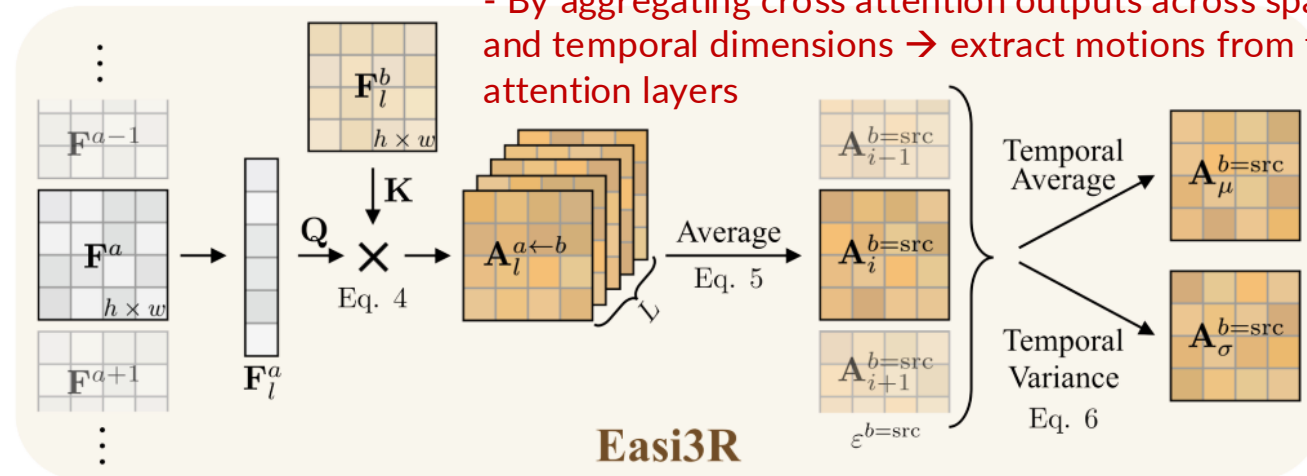
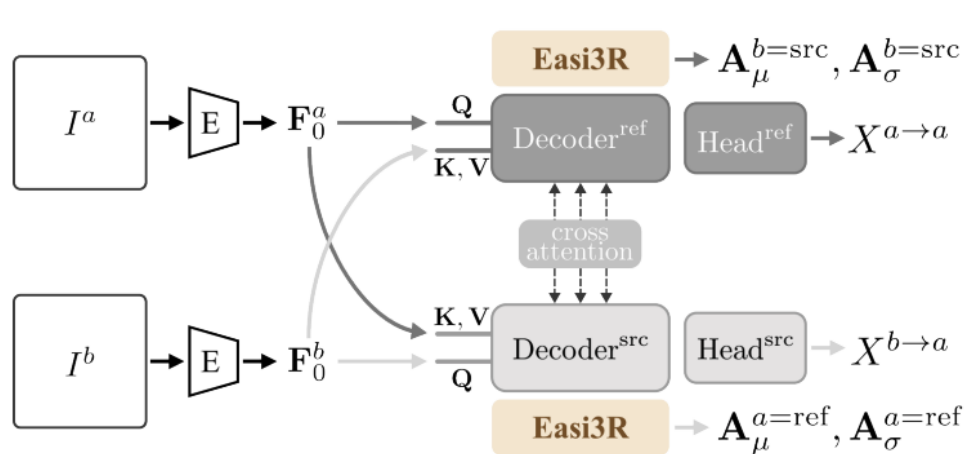
Tracking

Easi3R (ICCV'25)

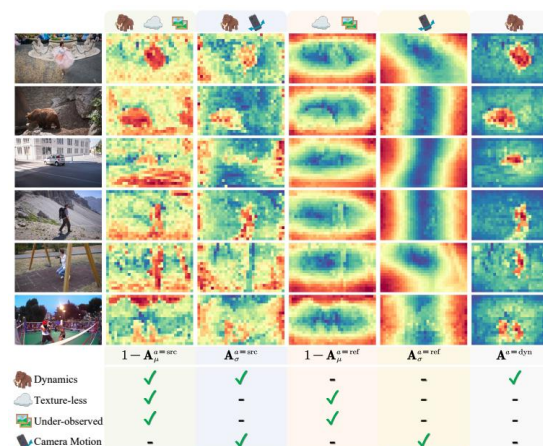
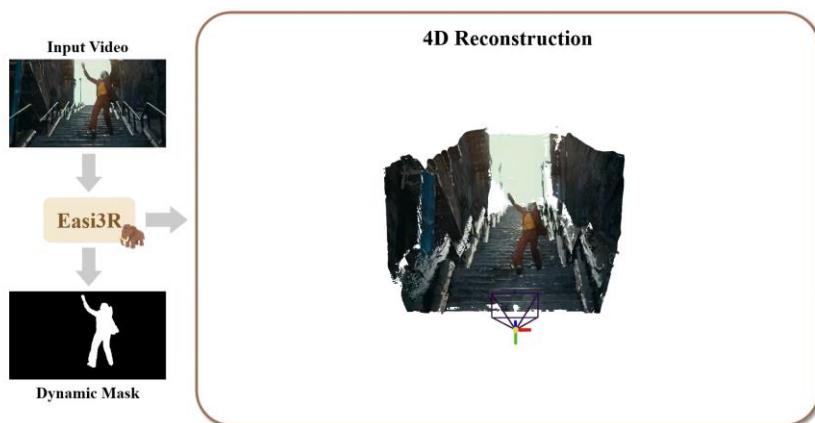
Easi3R: Estimating Disentangled Motion from DUST3R Without Training

Xingyu Chen^{1,2} Yue Chen^{1,2} Yuliang Xiu^{2,3} Andreas Geiger⁴ Anpei Chen^{2,4}
¹Zhejiang University ²Westlake University ³Max Planck Institute for Intelligent Systems
⁴University of Tübingen, Tübingen AI Center

A simple **training-free** approach adapting DUST3R for dynamic scenes.



- DUST3R implicitly learns rigid view transformations through its cross-attention layers;
- By aggregating cross attention outputs across spatial and temporal dimensions $\rightarrow$ extract motions from the attention layers



1. Extend DUST3R to dynamic scenes
2. Scene/human decomposition
3. Attention map disentanglement
4. Training-free control

Tracking

JOSH (submit to ICL)

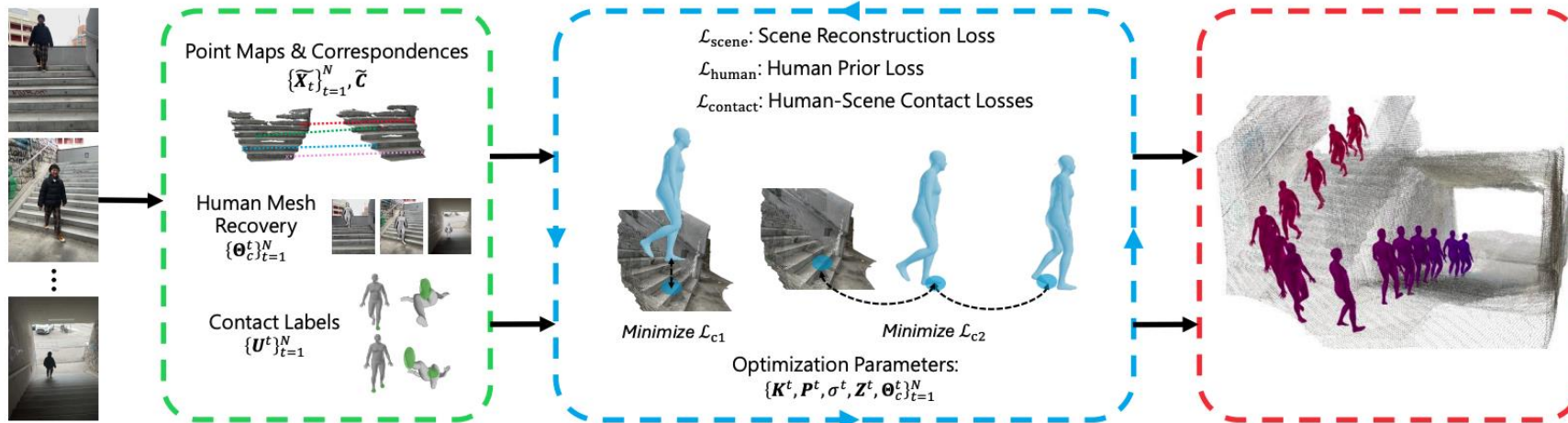
Methods	FPS↓	EMDB			SLOPER4D			RICH		
		WA-MPJPE↓	W-MPJPE↓	RTE%↓	WA-MPJPE↓	W-MPJPE↓	RTE%↓	WA-MPJPE↓	W-MPJPE↓	RTE%↓
SLAHMR	0.1	326.9	776.1	10.2	706.9	3401.0	12.6	132.2	237.1	6.4
WHAM	4.4	131.1	335.3	4.1	297.7	1272.3	10.5	108.4	196.1	4.5
TRAM	1.1	76.4	222.4	1.4	215.1	1285.3	3.0	127.8	238.0	6.0
JOSH ₃	0.8	68.9	174.7	1.3	120.0	438.3	1.8	102.6	184.3	3.4
JOSH3R	15.4	220.0	661.7	13.1	296.3	1148.4	10.4	190.4	334.9	6.3

Input Video

(a) Initialization

(b) Joint Optimization

4D Human-Scene Reconstruction



Joint Optimization for 4D Human-Scene Reconstruction in the Wild

Zhizheng Liu, Joe Lin, Wayne Wu, Bolei Zhou

University of California, Los Angeles

1. Multi-stage
2. Multi-person
3. Post-optimization



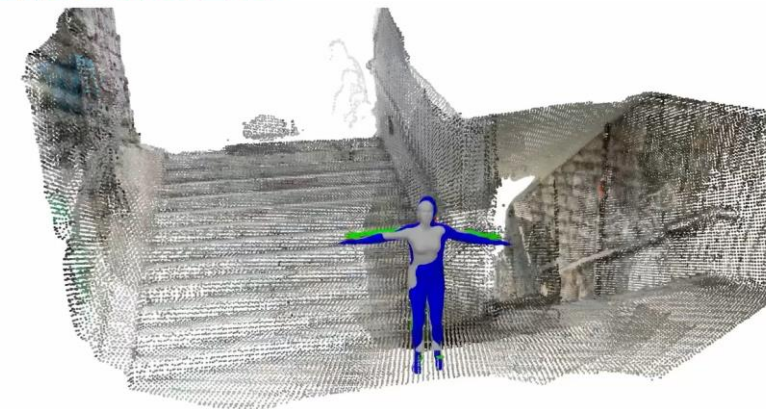
Input Web Video



4D Human-Scene Reconstruction



Input Video from EMDB



Ground Truth JOSH (ours) TRAM WHAM

*Dense scene reconstruction result is from JOSH as a reference, as no GT is available for EMDB and WHAM and TRAM cannot reconstruct the scene

Tracking

Human3R (arxiv'25)



Human3R: Everyone Everywhere All at Once

Yue Chen¹ Xingyu Chen^{1*} Yuxuan Xue² Anpei Chen¹ Yuliang Xiu^{1†} Gerard Pons-Moll^{2,3}

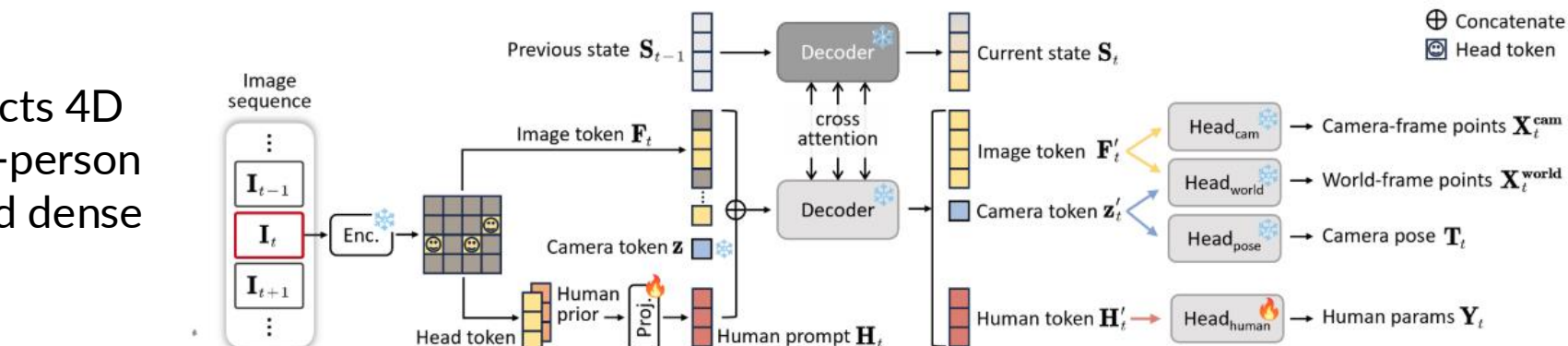
¹Westlake University ²University of Tübingen, Tübingen AI Center ³Max Planck Institute for Informatics

*Project Lead †Corresponding Author

Inference with One model, One stage; Training in One day using One GPU

Input: a stream of RGB images

Output: continuously reconstructs 4D human-scene in real time, multi-person meshes, camera parameters, and dense 3D geometry



Online Reconstruction  15 FPS, 8 GB GPU



Failure case (HOI)



1. CUT3R + Multi-HMR
2. Only fine-tune human parameters
3. Real-time (15fps)

Limitations

1. Penetration
2. HOI

Tracking

Uni3C (SigAsia'25)

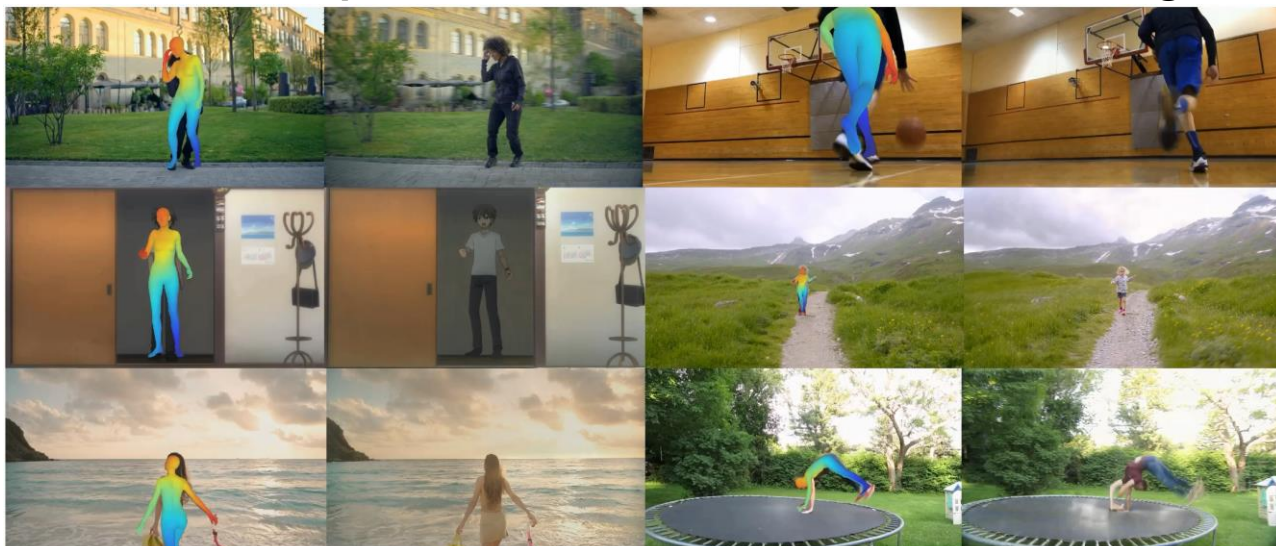
A unified 3D-enhanced framework for precise control of both camera and human motion in video generation

Input: single-view image

Output: controllable video generation (joint control of camera & human motion)

3D-enhanced:

- Input image -> Depth map -> Point cloud
- New cam param + Point cloud -> novel view image



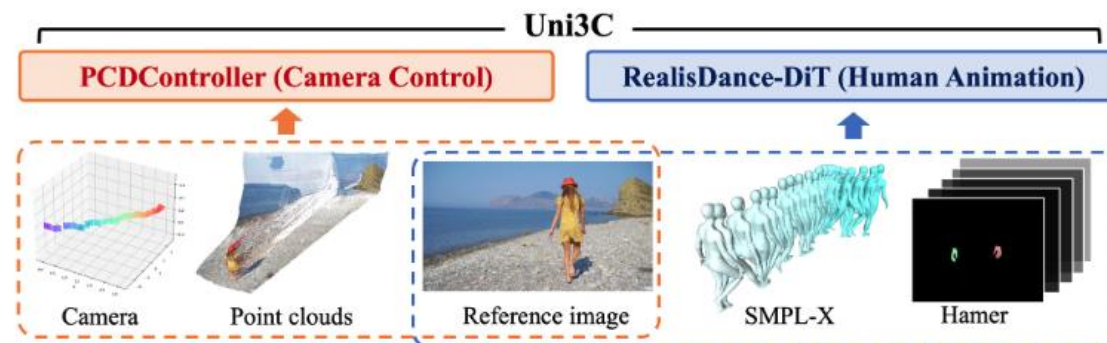
Unified camera and human motion control

Unifying Precisely 3D-Enhanced Camera and Human Motion Controls for Video Generation

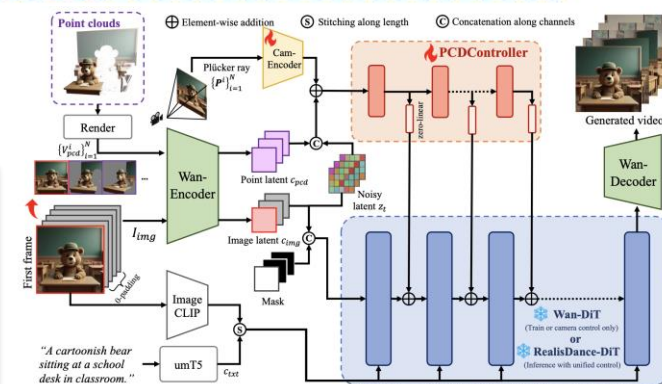
Siggraph Asia 2025

Chenjie Cao^{1,2,3}, Jingkai Zhou^{1,3}, Shikai Li^{1,3}, Jingyun Liang^{1,3}, Chaohui Yu^{1,3}, Fan Wang¹, Xiangyang Xue², Yanwei Fu^{2,4},

¹Alibaba DAMO Academy, ²Fudan University, ³Hupan Lab, ⁴Shanghai Innovation Institute



1. Use video generation model
2. Formulate as video generation
3. Large compute cost (64 H100, 40 hours)



RealisDance (arxiv'24)

<https://alibaba-damo-academy.github.io/Uni3C/>

Tracking

SHARE (arxiv'25, sig template with 4 pages)

SHARE: Scene-Human Aligned Reconstruction

Joshua Li
National Research Council Canada
University of Waterloo
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University of Waterloo
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Chang Shu
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Xue Bin Peng
Simon Fraser University
Canada
xbpeng@sfu.ca

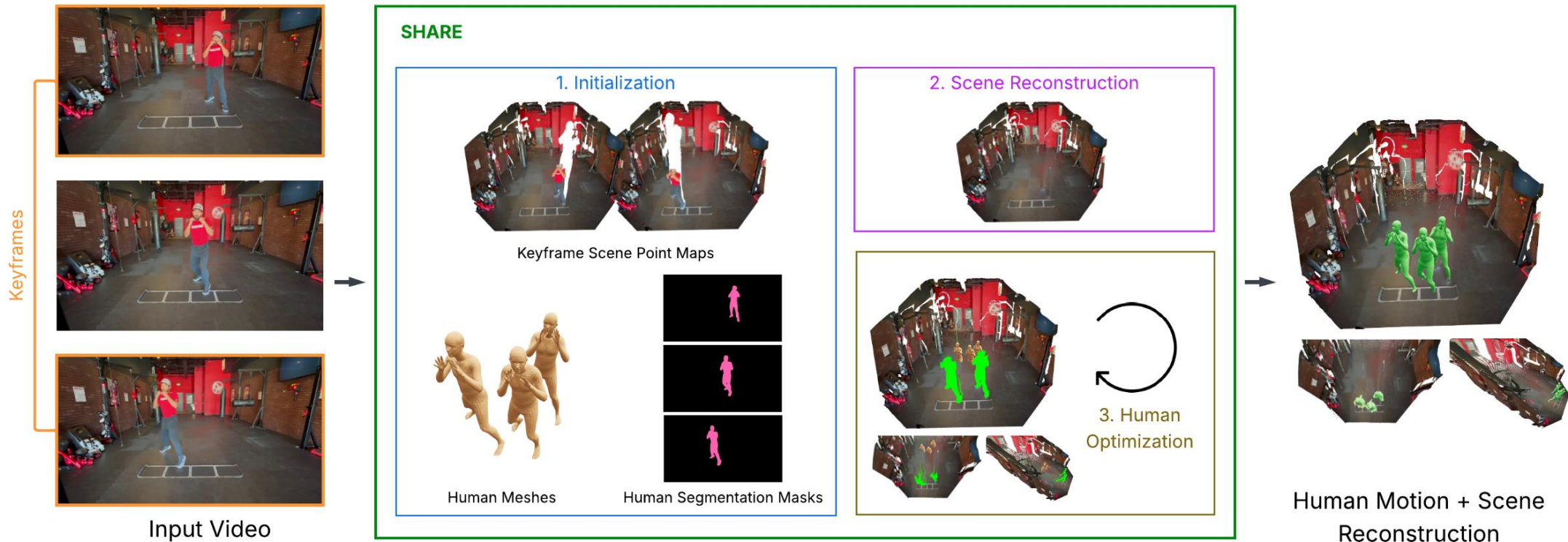
Pengcheng Xi
National Research Council Canada
Canada
pengcheng.xi@nrc-cnrc.gc.ca

Input: a video

Output: SMPL + scene point map

Multi-stage & optimization

1. TRAM + MoGe-2
2. Post optimization



Tracking

3D Human Pose/Motion Tracking

- Joint Human-Scene Tracking

- Global human motion estimation
- Generic scene reconstruction
- Interaction (scene/objects)
- Novel view synthesis, Generation (animation, motion transfer)
- Physics-aware

- Possible directions

- Physics-aware, simulation-ready
 - Terrain
 - Contact
- Human-object interaction
 - Rigid object
 - Deformable object
 - Articulated object
- Video generation/World models

Method	Venue	Note
HOSNeRF	ICCV'23	opt; dynamic NeRF
DUST3R	CVPR'24	ff; 3D GT; static (2 images)
VGGT	CVPR'25	ff; 3D GT; static
CUT3R	CVPR'25	ff; 3D GT; dynamic online
HSfM	CVPR'25	opt; static; single-frame, <u>multiview</u>
HAMSt3R	ICCV'25	ff; static; single-frame, <u>multiview</u>
<u>Easi3R</u>	ICCV'25	training-free; dynamic; (ext DUST3R)
Uni3C	SigAsia'25	video generation
JOSH3R	<u>arxiv'25</u>	opt; dynamic; (submit to ICLR26, 4666)
Human3R	<u>arxiv'25</u>	ff; dynamic online; (CUT3R+SMPL)
DA3	<u>arxiv'25</u>	ff; 3D GT; dynamic; (vs. VGGT)
SHARE	<u>arxiv'25</u>	opt; dynamic